Java MCQS

1) What will be the output of the following program?

```
public class IncrementDecrementQuiz
{
   public static void main(String[] args)
   {
      int i = 11;

      i = i++ + ++i;

      System.out.println(i);
   }
}
```

2) Guess the output of the following program?

```
public class IncrementDecrementQuiz
{
   public static void main(String[] args)
   {
      int a=11, b=22, c;

      c = a + b + a++ + b++ + ++a + ++b;

      System.out.println("a="+a);
      System.out.println("b="+b);
      System.out.println("c="+c);
   }
}
```

3) **What will be the output of the following program?**

```
public class IncrementDecrementQuiz
{
   public static void main(String[] args)
   {
      int a=1, b=2;

      System.out.println(--b - ++a + ++b - --a);
   }
}
```

4)

```
int[] numbers = {1, 2, 3};
for (int num : numbers) {
   num = num * 2;
}
// What are the values in the 'numbers' array after the loop?
```

5)

```
public class Calculator
{
   int num = 100;
   public void calc(int num)  { this.num = num * 10;  }
   public void printNum()     { System.out.println(num); }

   public static void main(String[] args)
   {
      Calculator obj = new Calculator();
      obj.calc(2);
      obj.printNum();
   }
}
```

6) Find the output:

```
class Alpha
{
   public String type = "a ";
   public Alpha() {  System.out.print("alpha "); }
}

public class Beta extends Alpha
{
   public Beta()  {  System.out.print("beta ");  }

   void go()
   {
      type = "b ";
      System.out.print(this.type + super.type);
   }

   public static void main(String[] args)
   {
      new Beta().go();
   }
}
```

7) Which of the following access modifiers makes a member accessible only within its own class?
a) public
b) protected
c) private
d) default (package-private)

8) What is the default access modifier for a class member if no explicit access modifier is specified?
a) private
b) public
c) protected
d) default (package-private)

9) A protected member of a class is accessible:
a) Only within the class where it is declared.
b) Within the same package and by subclasses in any package.
c) From any class, regardless of its package.
d) Only by subclasses within the same package.

10) What is the primary purpose of the final keyword in Java?
a) To make a class abstract
b) To prevent inheritance, method overriding, and variable modification
c) To define a static method
d) To handle exceptions

11)
```
class Main {
 public static void main(String args[]){
   final int i;
   i = 20;
   System.out.println(i);
 }
}
```

12) Find the output
```
class Base {
  public final void show() {
      System.out.println("Base::show() called");
    }
}
class Derived extends Base {
    public void show() {
      System.out.println("Derived::show() called");
    }
}
public class Main {
```

```
    public static void main(String[] args) {
        Base b = new Derived();;
        b.show();
    }
}
```

13) Find the output:

```
public class NewClass {
     private int x;
    public void Example() { System.out.println(x); }
   public static void main(String args[])
   {
      NewClass obj = new NewClass();
      obj.Example();
   }
  }
```

14)

```
int x = 10, y = 20;
System.out.println(x > y ? x : y);
```

15) What does the keyword `static` mean in Java?

- A) The method cannot be overridden
- B) The method belongs to the class, not to an object
- C) The method can be accessed from outside the class
- D) The method is synchronized

16) In which package are the classes `System` and `String` defined?

- A) `java.util`
- B) `java.io`
- C) `java.lang`
- D) `java.text`